

KUBERNETES RELEASE WEBINAR

Kubernetes v1.37

Garhwal

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The Release Theme

Garhwal is a Himalayan region of Uttarakhand, India. Its landscape, craft, and wildlife shape the v1.37 artwork.

Terraced fields rise toward the snowy peaks, each layer supported by the one below. The river gathers mountain streams, reflecting contributions from many SIGs flowing into one project.

The woven ringaal frame represents strength through interlinked work. Deodar trees, the Himalayan monal, and red buransh flowers root the design in Garhwal.

Watch for 37 seconds: the river flows, the flags move, and the landscape reveals an Easter egg.

<https://kubernetes.io/blog/2026/08/26/kubernetes-v1-37-release/>



Release at a glance

67

enhancements

16

Stable

23

Beta

27

Alpha

1

deprecation / removal

15 weeks

Release cycle

18 May- 26 Aug 2026

Cycle dates

Up to 212 companies

Community reach

1,754 individuals

Contributors



RELEASE HIGHLIGHTS

SIG API Machinery & etcd

Control plane scalability, admission configuration, and more efficient etcd access.



Resilient watchcache initialization

SIG API Machinery

WatchCacheInitializationPostStartHook graduates to Stable and is locked on. During cache warm-up, the API server bounds requests delegated to etcd and rejects expensive requests with HTTP 429.

This protects etcd and API Priority and Fairness during API server restarts. Clients should honor Retry-After and use exponential backoff. The initialization hook was enabled by default in v1.36; resilient watch-cache initialization itself was already Stable in v1.34.

STABLE

<https://www.k8s.dev/resources/keps/4568/>

Enabled by default in v1.37.

Storage Version Migrator

SIG API Machinery

Storage Version Migrator graduates to Stable with `storagemigration.k8s.io/v1`. An in-tree controller rewrites stored objects to the current storage version and encryption configuration.

Create a `StorageVersionMigration` resource to migrate built-in resources or CRDs declaratively. This replaces ad hoc rewrite scripts and supports storage-version changes and encryption-at-rest rotation. The stable capability is enabled by default in v1.37.

STABLE

<https://www.k8s.dev/resources/keps/4192/>

Stable API:
`storagemigration.k8s.io/v1`

Manifest based admission control

SIG API Machinery

AdmissionConfiguration can load webhook and CEL policy manifests from staticManifestsDir. Policies apply from API server startup and remain available when etcd is unavailable.

Use disk-backed policies to protect admission resources themselves and establish a startup baseline. The API server watches for manifest changes and reloads valid updates; an invalid update retains the previous configuration. Manage and distribute these files consistently across API servers.

BETA

<https://www.k8s.dev/resources/keps/5793/>

A new beta configuration path.

etcd RangeStream

SIG etcd

EtcdRangeStream streams large etcd range responses in adaptive chunks, reducing peak memory during watch-cache initialization. A pinned MVCC revision preserves a consistent snapshot across chunks.

Enabled by default in the API server, this beta requires etcd 3.7+ for streaming. Older servers automatically fall back to unary Range on Unimplemented. It also serves direct GetList requests when WatchList is disabled; compaction during a stream triggers a retry.

BETA

<https://www.k8s.dev/resources/keps/5966/>

Enabled by default; requires etcd 3.7+.

RELEASE HIGHLIGHTS

SIG Autoscaling

Scaling workloads all the way down when external demand disappears.



Horizontal Pod Autoscaler scale to zero

SIG Autoscaling

Set `spec.minReplicas: 0` to let an HPA scale idle workloads to zero using object or external metrics. The beta capability is enabled by default in v1.37.

For example, a queue consumer can stop when the queue is empty and resume when demand rises. CPU and memory metrics cannot activate a workload with no running pods. The `ScaledToZero` condition distinguishes HPA-driven zero replicas from manual deactivation.

BETA

<https://www.k8s.dev/resources/keps/2021/>

Enabled by default in v1.37.

RELEASE HIGHLIGHTS

SIG Scheduling

Coordinated placement for workloads whose pods need to start together.



Gang scheduling

SIG Scheduling

The Workload API and PodGroup support all-or-nothing scheduling for related pods. Gang scheduling reaches Beta, helping workloads obtain the resources needed to make progress together.

A distributed training job can wait until its worker group fits instead of occupying resources with an unusable subset. Beta adds workload-aware preemption and PodGroup queueing, and addresses livelock when multiple workloads compete.

BETA

<https://www.k8s.dev/resources/keps/4671/>

Graduated to Beta in v1.37.

DRA extended resources

SIG Scheduling

DRA extended-resource support graduates to Stable. A DeviceClass can expose an extended resource name, allowing existing pod resource requests to use Dynamic Resource Allocation.

For example, a pod requesting `abc.example/gpu: 3` can obtain devices through DRA without adding a ResourceClaim manifest. Administrators configure the DeviceClass mapping and a compatible DRA driver, while application teams retain familiar resource requests.

STABLE

<https://www.k8s.dev/resources/keps/5004/>

Graduated to Stable in v1.37.

RELEASE HIGHLIGHTS

SIG Node

New workload lifecycle and memory isolation capabilities at the node.



Pod level checkpoint and restore

SIG Node

Alpha CRI CheckpointPod and RestorePod RPCs let the kubelet request pod-level checkpoint and restore from a compatible container runtime. Runtime implementation of both operations is required.

This provides a foundation for workload migration and recovery by preserving running state. Enable the alpha capability only with a runtime that supports the new RPCs, and validate workload compatibility and restore behavior before depending on it.

ALPHA

<https://www.k8s.dev/resources/keps/5823/>

Alpha; requires explicit
enablement.

Memory QoS

SIG Node

MemoryQoS reaches Beta and is enabled by default on Linux with cgroups v2. It uses memory.min, memory.low, and memory.high to protect memory and control reclaim or throttling.

Memory requests inform protection under contention. Kubelet memoryReservationPolicy and memoryThrottlingFactor control reservation and throttling behavior. Defaults avoid unexpected throttling on upgrade; evaluate workload requests and node pressure when tuning these settings.

BETA

<https://www.k8s.dev/resources/keps/2570/>

Enabled by default in v1.37.

Declared node features

SIG Node

Node declared features graduate to Stable. Kubelets publish supported capabilities in `status.declaredFeatures`, making node feature support explicit during mixed-version cluster upgrades.

Scheduling and other consumers can use declared capabilities instead of inferring support solely from a node version. The kubelet derives these declarations from static configuration and feature gates at startup; restart the kubelet when relevant configuration changes.

STABLE

<https://www.k8s.dev/resources/keps/5328/>

Derived at kubelet startup

RELEASE HIGHLIGHTS

SIG CLI

KYAML - an alternative to standard YAML.



KYAML input and output

SIG CLI

KYAML output graduates to Stable: use `kubectl get -o kyaml` for a safer, YAML-compatible representation. KYAML support is conformance-tested in v1.37.

Any `kubectl` version can parse KYAML because it remains valid YAML. Existing manifests and pipelines do not need conversion. The output format progressed from Alpha in v1.34 to Beta in v1.35 and now provides a stable option for configuration tooling.

STABLE

<https://www.k8s.dev/resources/keps/5295/>

Graduated to Stable in v1.37.

RELEASE HIGHLIGHTS

SIG Instrumentation

A stable resource metrics API and native histogram support for Kubernetes.



metrics.k8s.io API stability

SIG Instrumentation

The resource metrics API graduates to metrics.k8s.io/v1 after nearly nine years in Beta. It provides CPU and memory usage for nodes and pods, supporting HPA and kubectl top.

Clients gain a stable API contract for resource usage. The v1beta1 endpoint remains supported during the transition under Kubernetes deprecation policy; migration to v1 will continue in future releases. Check API-version support when updating metrics clients and integrations.

STABLE

<https://www.k8s.dev/resources/keps/5207/>

Graduated to Stable in v1.37.

Native histograms

SIG Instrumentation

Native histogram support reaches Beta. Dynamic exponential buckets capture metric distributions with greater flexibility than fixed bucket boundaries, supporting more precise latency analysis.

Enable the NativeHistograms feature gate and use Prometheus protobuf scraping to collect native and classic histograms together. During rollout, validate Prometheus ingestion, PromQL queries, dashboards, and alerting; classic histograms remain available as a compatibility path while observability tooling transitions to native histograms.

BETA

<https://www.k8s.dev/resources/keps/5808/>

NativeHistograms feature gate

RELEASE HIGHLIGHTS

SIG Storage

SELinux volume relabeling



SELinux volume handling

SIG Storage

SELinuxMount and SELinuxChangePolicy graduate to Stable and default to true. Supported volumes and CSI drivers can apply an SELinux context at mount time instead of recursively relabeling every file.

Volumes can opt-in via `-o context=<label>` CSI drivers opt in with `spec.seLinuxMount: true`. Different SELinux labels on a shared volume on one node can conflict and cause the pods to fail to start. To retain the old behavior, set pod `spec.seLinuxChangePolicy: Recursive`. The gates remain disableable in v1.37 and have no effect without SELinux. However, this behavior becomes the default in v1.38.

STABLE

<https://www.k8s.dev/resources/keps/1710/>

Drivers must advertise support.

Other SIG highlights

SIG API Machinery

Storage Version Migrator reaches Stable.

SIG Auth

Pod certificates and ClusterTrustBundles reach Stable.

SIG Storage

PVC last-used tracking reaches Beta.

SIG Apps

StatefulSet Recreate update strategy begins in Alpha.

SIG Scheduling / Node

Dynamic Resource Allocation capabilities continue graduating to Stable.

Deprecations and compatibility changes

Paths to review before future releases remove or disable them.



SIG Network: deprecation milestones

SIG Network

kube-dns

Deprecated in v1.37. The project expects no new kube-dns packages after v1.40. CoreDNS remains the recommended DNS server.

kube-proxy IPVS mode

Now emits a startup warning. The release plan expects it to become disabled by default in v1.40 and removed in v1.43.

Review cluster DNS and service proxy choices before later releases close these migration paths.

SIG CLI and SIG Node: compatibility changes

SIG CLI · `kubectl run`

The `--filename` and `-f` flags are deprecated.

SIG Node · Static Pods

Static Pods cannot reference Secrets or ConfigMaps.

SIG Node · cgroup v1

The project continues toward removal. `failCgroupV1` has defaulted to `true` since v1.35.

BEHIND THE RELEASE

Release Team

What the v1.37 cycle taught us and what changes for v1.38.



v1.38: a longer implementation window

Release Team

Lesson from v1.37

A record number of Code Freeze exceptions highlighted limited reviewer and approver capacity near the deadline.

Current v1.38 cycle

Earlier team selection and an earlier cycle start create more time for implementation and review.

About two more weeks from Enhancements Freeze to Code Freeze

More time to prepare, review, and merge enhancements.

v1.37 cycle retrospective

Release Team

Release scope

67 enhancements

16 Stable · 23 Beta · 27 Alpha

1 deprecation or removal

Community and timing

15-week release cycle

18 May - 26 August 2026

Up to 212 companies and 1,754 individuals

Thank you to every contributor, reviewer, tester, documentarian, SIG lead, and release-team volunteer.

Resources and ways to participate

Release announcement

<https://kubernetes.io/blog/2026/08/26/kubernetes-v1-37-release/>

Release notes

<https://github.com/kubernetes/kubernetes/blob/master/CHANGELOG/CHANGELOG-1.37.md>

Release directory

<https://github.com/kubernetes/sig-release/tree/master/releases/release-1.37>

Join the community

<https://kubernetes.io/community>

Choose a SIG, join #sig-release on Kubernetes Slack, or follow the release-team mailing list.

Thank you

Questions and discussion

